BRAZILIAN E-COMMERCE : Business Insights & Recommendations

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Business Problem

E-commerce companies must balance growth, operations, and customer satisfaction

Need to understand customer behavior and delivery performance

Goal: use data to improve business decisions

Project Objective

- Analyze sales, customers, and operations
- Identify key drivers of revenue and satisfaction
- Provide actionable recommendations

Tools & Methodology

SQL

data extraction & joins

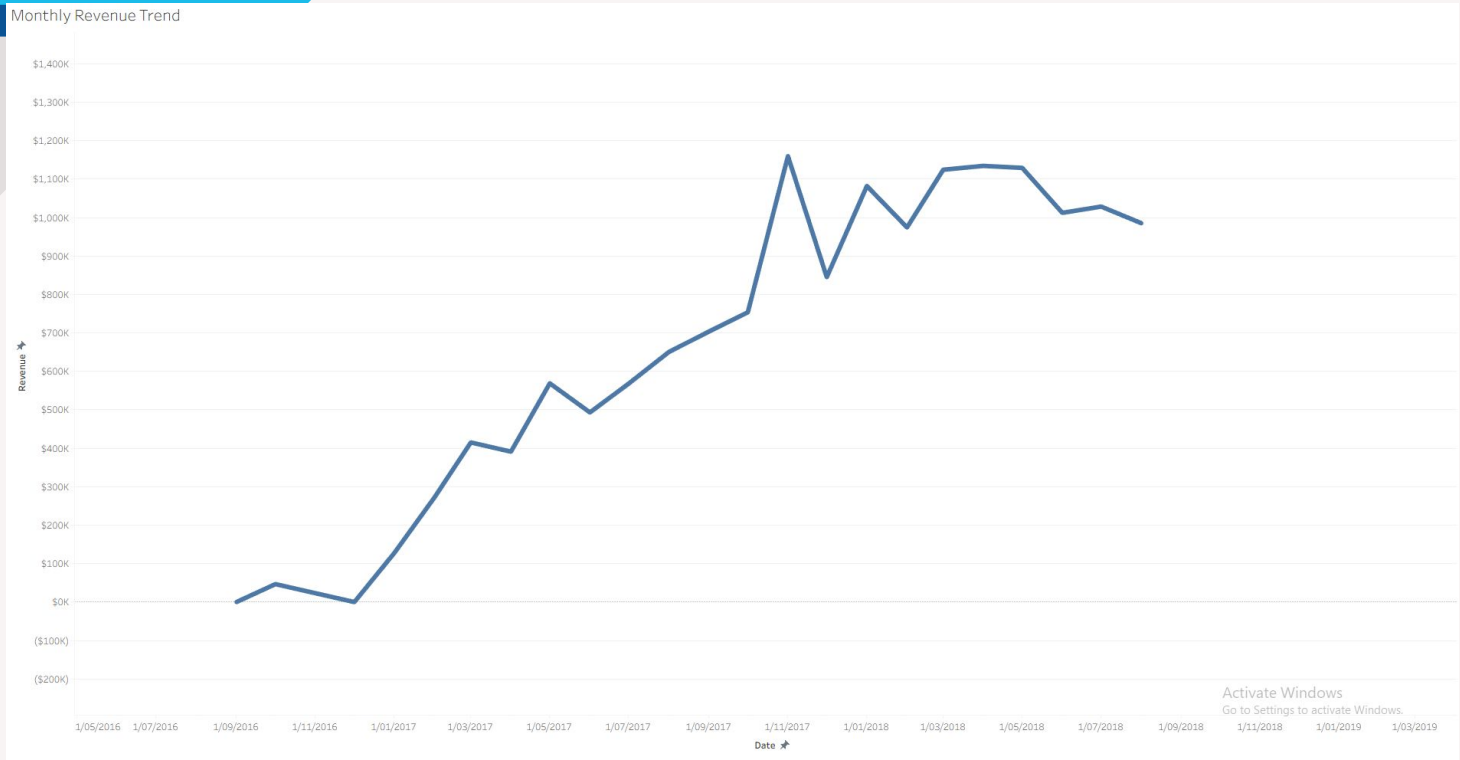
Python (pandas)

cleaning & feature engineering

Tableau

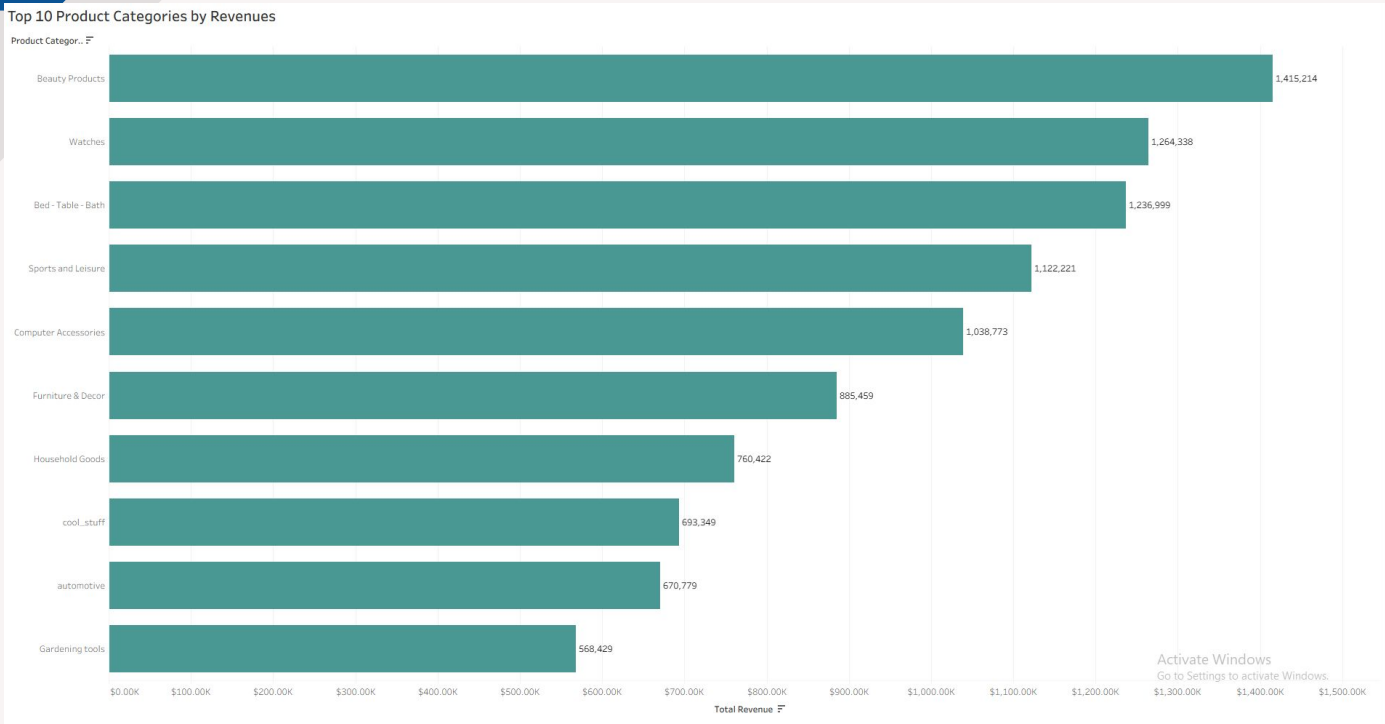
visualization

Revenue Trend



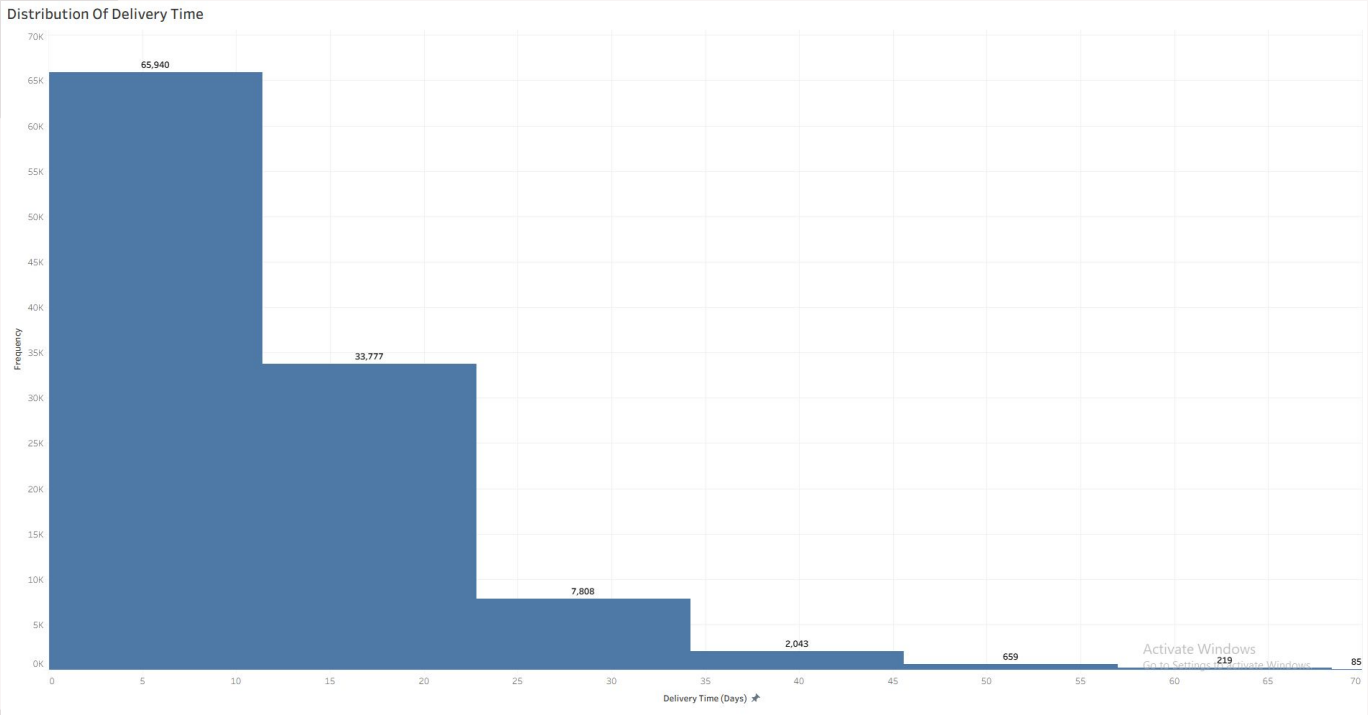
- Revenue experienced rapid growth in 2017 followed by stabilization, suggesting the business reached a maturity phase where growth strategies must shift from acquisition to retention and optimization.

Top Product Categories



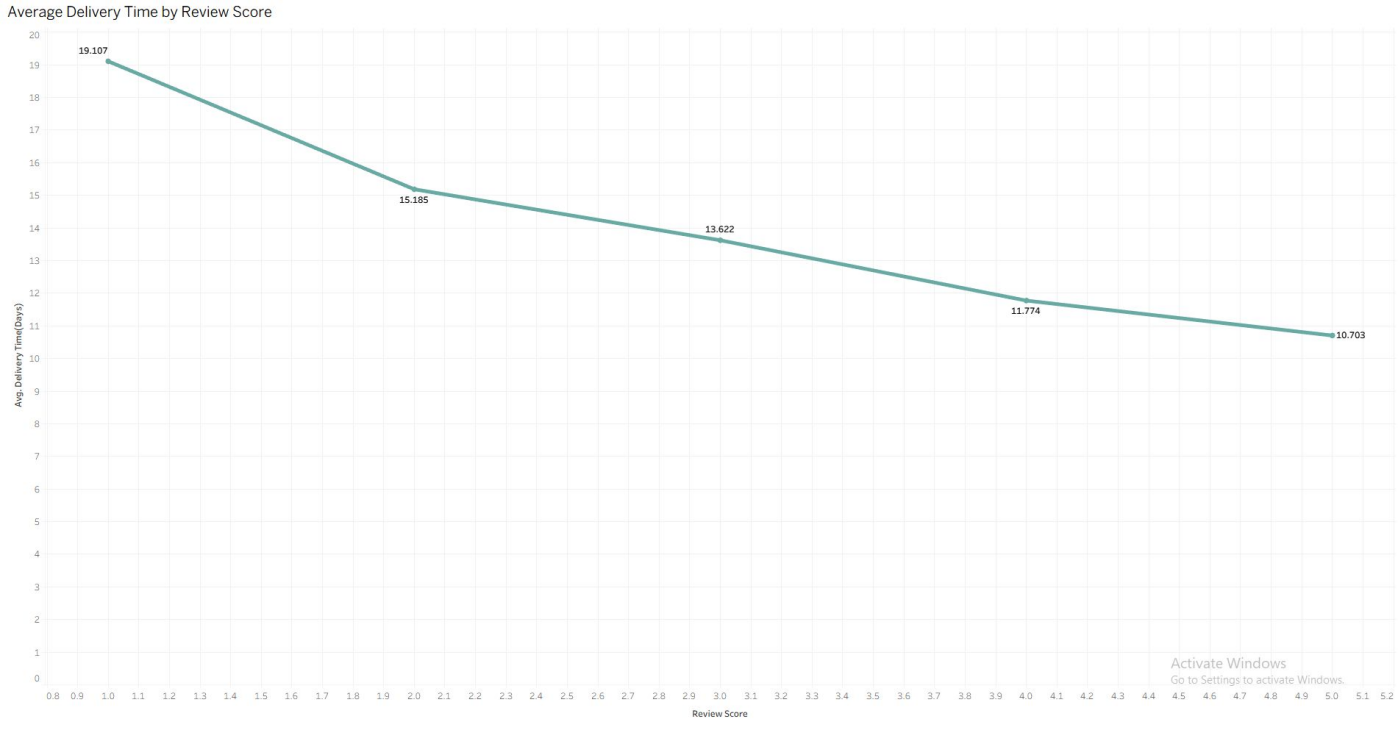
- Revenue is heavily concentrated in a small number of product categories, indicating a dependency risk but also an opportunity to double down on high-performing segments.

Delivery Time Distribution



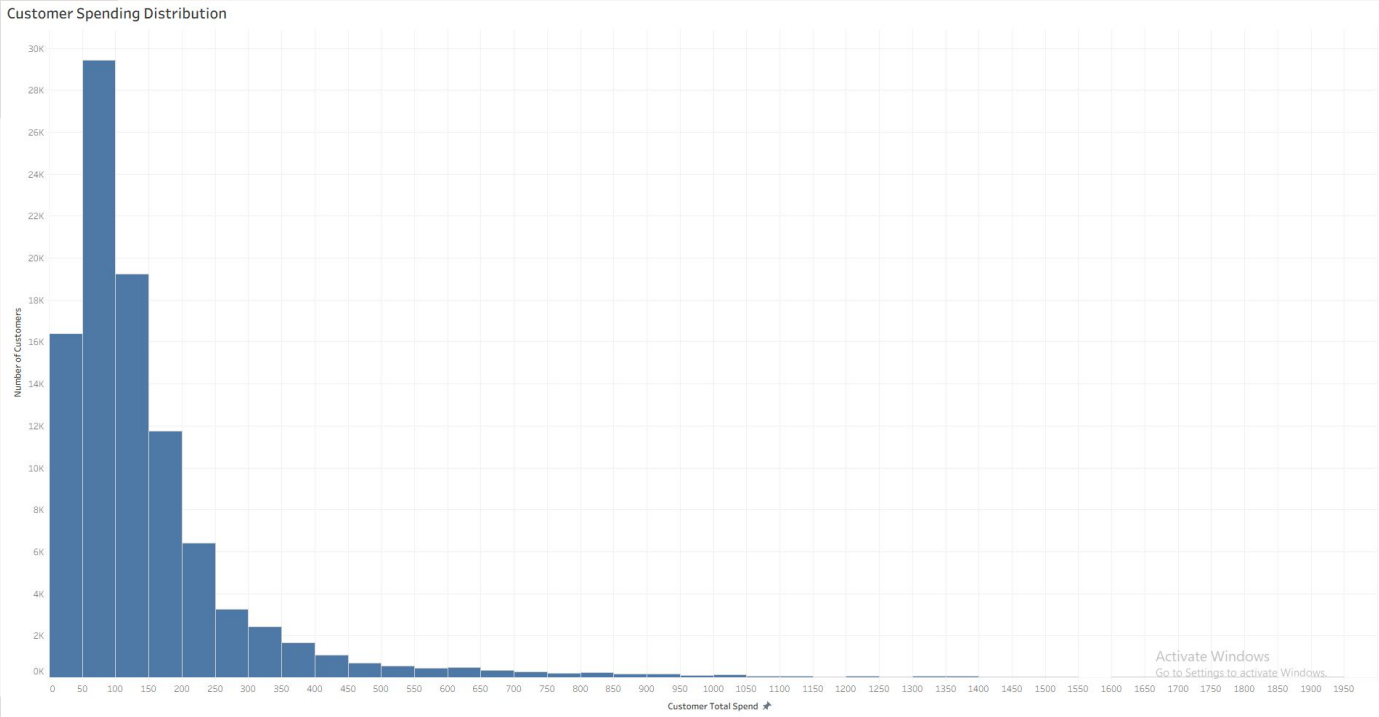
- Delivery times are highly variable, with significant outliers, highlighting inefficiencies in logistics that may negatively impact customer experience.

Delivery Time vs Customer Satisfaction



- Delivery speed has a direct and measurable impact on customer satisfaction, with longer delivery times consistently associated with lower review scores.

Customer Spending Distribution



- Customer spending follows a highly skewed distribution, where a small segment of high-value customers contributes disproportionately to total revenue.

Key Insights

Revenue showed strong growth in 2017 followed by a clear stabilization in 2018, suggesting the business is transitioning from a high-growth phase to a more mature stage where optimization and retention become more important.

Sales are highly concentrated in a small number of product categories, meaning a large portion of revenue depends on a few key segments. This highlights both an opportunity to focus on top performers and a potential risk if demand shifts.

Delivery time plays a major role in customer satisfaction, with longer delivery times consistently leading to lower review scores, showing that logistics performance directly impacts customer experience.

Customer spending is unevenly distributed, where a small group of high-value customers generates a significant share of total revenue, making them critical for long-term business growth.

Business Recommendations

- Optimize last-mile delivery operations in high-delay regions to reduce variability in delivery times
- Implement customer retention strategies targeting high-value segments (loyalty programs, personalized offers)
- Prioritize inventory and marketing investment in top-performing product categories
- Establish KPIs to continuously monitor delivery performance and customer satisfaction

Business Impact

1. Improving delivery time
2. ↑ customer satisfaction
3. ↑ repeat purchases
4. Focusing on top customers
5. ↑ revenue efficiency
6. ↑ ROI

THANKS !